INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL PRACTICAL & LABORATORY MANUAL â¢ DEPARTMENT OF CSE

Laboratory Course: [BCS-354] Python Lab

Programme: Bachelor of Technology in Computer Science & Engineering **Manual Type: LAB MANUAL**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Prof. Neha Gupta (Assistant**

Official Laboratory Manual: Python Programming, OOP & Data Analysis

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1. Python Laboratory Objectives

- Learning Objective: Fluency in Python 3 syntax, functional paradigms, OOP, and data packages.
- IDE / Environment: Python 3.10+, VS Code / PyCharm / Jupyter Notebook.
- Assessment: Program correctness, coding standards (PEP 8), exception handling, and lab viva.

2. Prescribed Laboratory Exercises

- Exercises 1-4: Datatypes, operator precedence, list manipulation methods, dictionary methods.
- Exercises 5-8: Tuples and sets, control structures, custom functions with keyword parameters.
- Exercises 9-12: String algorithms, text file word counting, OOP inheritance, and NumPy matrix multiplication.